

## THOMASVILLE 2 S, AL, USA

WMO: 721551

Lat: 31.8814N

Lon: 87.7367W

Elev: 107

StdP: 100.05

Time Zone: -6.00 (NAC)

Period: 08-19

WBAN: 23802

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |            |            |            |            | Coldest Month WS/MCDB |             |            |             | MCWS/PCWD to 99.6% DB |            | WSF        |
|---------------|-------------|-------------|-------------------------------|------------|------------|------------|------------|------------|-----------------------|-------------|------------|-------------|-----------------------|------------|------------|
|               |             |             | 99.6%                         |            |            | 99%        |            |            | 0.4%                  |             | 1%         |             |                       |            |            |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB       | DP         | HR         | MCDB       | WS                    | MCDB        | WS         | MCDB        | MCWS                  | PCWD       |            |
| (a)<br>1      | (b)<br>-5.1 | (c)<br>-3.0 | (d)<br>N/A                    | (e)<br>N/A | (f)<br>N/A | (g)<br>N/A | (h)<br>N/A | (i)<br>N/A | (j)<br>6.6            | (k)<br>16.9 | (l)<br>5.8 | (m)<br>13.9 | (n)<br>1.7            | (o)<br>350 | (p)<br>N/A |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |            |             |            |             |            | Evaporation WB/MCDB |            |            |            |            |            | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|------------|-------------|------------|-------------|------------|---------------------|------------|------------|------------|------------|------------|----------------------|------------|
|               |                        | 0.4%            |            | 1%          |            | 2%          |            | 0.4%                |            | 1%         |            | 2%         |            |                      |            |
|               |                        | DB              | MCWB       | DB          | MCWB       | DB          | MCWB       | WB                  | MCDB       | WB         | MCDB       | WB         | MCDB       | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>10.6            | (c)<br>34.6     | (d)<br>N/A | (e)<br>33.5 | (f)<br>N/A | (g)<br>32.5 | (h)<br>N/A | (i)<br>N/A          | (j)<br>N/A | (k)<br>N/A | (l)<br>N/A | (m)<br>N/A | (n)<br>N/A | (o)<br>1.9           | (p)<br>350 |

| Dehumidification DP/MCDB and HR |            |            |            |            |            |            |            |            | Enthalpy/MCDB |            |            |            |            |            | Extreme<br>Max WB |
|---------------------------------|------------|------------|------------|------------|------------|------------|------------|------------|---------------|------------|------------|------------|------------|------------|-------------------|
| 0.4%                            |            |            | 1%         |            |            | 2%         |            |            | 0.4%          |            | 1%         |            | 2%         |            |                   |
| DP                              | HR         | MCDB       | DP         | HR         | MCDB       | DP         | HR         | MCDB       | Enth          | MCDB       | Enth       | MCDB       | Enth       | MCDB       |                   |
| (a)<br>N/A                      | (b)<br>N/A | (c)<br>N/A | (d)<br>N/A | (e)<br>N/A | (f)<br>N/A | (g)<br>N/A | (h)<br>N/A | (i)<br>N/A | (j)<br>N/A    | (k)<br>N/A | (l)<br>N/A | (m)<br>N/A | (n)<br>N/A | (o)<br>N/A | (p)<br>N/A        |

## Extreme Annual Design Conditions

| Extreme Annual WS |         |         | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |  |  |
|-------------------|---------|---------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|--|--|
|                   |         |         | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |  |  |
| 1%                | 2.5%    | 5%      | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      |  |  |
| (a) 5.9           | (b) 5.2 | (c) 4.5 | (e) -8.3                   | (f) 35.7 | (g) 2.7            | (h) 1.7 | (i) -10.3                                          | (j) 37.0 | (k) -11.9  | (l) 38.0 | (m) -13.4  | (n) 38.9 | (o) -15.4  | (p) 40.2 |  |  |
|                   |         |         | DB                         |          |                    |         |                                                    |          |            |          |            |          |            |          |  |  |
|                   |         |         | WB                         |          |                    |         |                                                    |          |            |          |            |          |            |          |  |  |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 18.0   | 7.8   | 10.9  | 14.1  | 17.9  | 21.8  | 25.8  | 26.6  | 26.3  | 24.3  | 18.2  | 12.3  | 10.1  |
|                                                                               | DBStd       | 7.77   | 5.99  | 5.63  | 5.00  | 3.65  | 3.13  | 1.97  | 1.51  | 1.77  | 2.66  | 4.27  | 4.68  | 5.49  |
|                                                                               | HDD10.0     | 291    | 115   | 54    | 21    | 1     | 0     | 0     | 0     | 0     | 0     | 2     | 28    | 71    |
|                                                                               | HDD18.3     | 1261   | 328   | 216   | 147   | 53    | 11    | 0     | 0     | 0     | 1     | 56    | 188   | 262   |
|                                                                               | CDD10.0     | 3219   | 47    | 78    | 147   | 236   | 367   | 473   | 514   | 504   | 428   | 256   | 97    | 73    |
|                                                                               | CDD18.3     | 1147   | 2     | 7     | 14    | 38    | 119   | 223   | 256   | 246   | 179   | 51    | 7     | 5     |
|                                                                               | CDH23.3     | 10052  | 5     | 26    | 102   | 350   | 1026  | 2116  | 2312  | 2116  | 1487  | 466   | 38    | 9     |
|                                                                               | CDH26.7     | 4092   | 0     | 1     | 9     | 53    | 339   | 969   | 1048  | 942   | 618   | 110   | 3     | 0     |
| Wind                                                                          | WSAvg       | 1.8    | 2.1   | 2.5   | 2.6   | 2.1   | 1.6   | 1.6   | 1.5   | 1.5   | 1.3   | 1.6   | 1.6   | 2.1   |
| Precipitation                                                                 | PrecAvg     | 1500   | 145   | 131   | 169   | 108   | 110   | 120   | 155   | 112   | 105   | 86    | 117   | 138   |
|                                                                               | PrecMax     | 2115   | 309   | 364   | 390   | 305   | 306   | 322   | 404   | 252   | 341   | 264   | 286   | 462   |
|                                                                               | PrecMin     | 1015   | 28    | 32    | 32    | 12    | 0     | 6     | 18    | 26    | 2     | 0     | 22    | 28    |
|                                                                               | PrecStd     | 254    | 64    | 69    | 92    | 67    | 67    | 74    | 78    | 63    | 77    | 72    | 68    | 77    |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 24.0  | 26.0  | 28.0  | 29.2  | 32.9  | 35.6  | 35.3  | 35.5  | 34.9  | 30.8  | 26.8  | 24.9  |
|                                                                               |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                               | 2%          | DB     | 21.8  | 23.9  | 26.2  | 28.2  | 31.3  | 34.4  | 34.0  | 34.2  | 33.2  | 29.4  | 24.2  | 22.8  |
|                                                                               |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                               | 5%          | DB     | 19.7  | 21.8  | 24.2  | 27.1  | 30.0  | 33.1  | 32.9  | 32.8  | 31.8  | 27.9  | 22.6  | 21.2  |
|                                                                               |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                               | 10%         | DB     | 17.5  | 20.0  | 22.3  | 25.6  | 28.8  | 31.7  | 31.9  | 31.5  | 30.5  | 26.2  | 20.8  | 19.5  |
|                                                                               |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                               |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |       |
|                                                                               | 2%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                               |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |       |
|                                                                               | 5%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                               |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |       |
|                                                                               | 10%         | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                               |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |       |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 11.9  | 12.1  | 12.7  | 13.0  | 12.2  | 11.4  | 10.6  | 10.5  | 11.3  | 13.3  | 13.4  | 10.8  |
|                                                                               |             | MCDBR  | 12.3  | 12.3  | 13.2  | 13.9  | 13.8  | 13.1  | 12.0  | 12.3  | 13.2  | 14.4  | 14.0  | 10.9  |
|                                                                               |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                               | 5% WB       | MCDBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                               |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        |        | 0.321 | 0.337 | 0.361 | 0.388 | 0.434 | 0.461 | 0.523 | 0.514 | 0.439 | 0.374 | 0.344 | 0.326 |
|                                                                               | taud        |        | 2.544 | 2.502 | 2.438 | 2.355 | 2.264 | 2.236 | 2.077 | 2.098 | 2.289 | 2.447 | 2.482 | 2.539 |
|                                                                               | Ebn at Noon |        | 906   | 922   | 920   | 903   | 859   | 831   | 779   | 781   | 830   | 868   | 869   | 880   |
|                                                                               | Edh at Noon |        | 84    | 96    | 110   | 124   | 137   | 141   | 164   | 159   | 126   | 100   | 89    | 81    |
| All-Sky Solar<br>Radiation                                                    | RadAvg      |        | 2.81  | 3.37  | 4.53  | 5.63  | 6.19  | 6.17  | 5.94  | 5.55  | 5.01  | 4.27  | 3.21  | 2.44  |
|                                                                               | RadStd      |        | 0.21  | 0.45  | 0.40  | 0.46  | 0.40  | 0.49  | 0.37  | 0.34  | 0.36  | 0.51  | 0.30  | 0.24  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (43 neighbors) |  | +0.45 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +148    | +96     |

Nomenclature: See separate page